

Alumnex Connect

Budget & Pricing Estimation

A comprehensive analysis of all infrastructure, APIs, and services required to publish Alumnex Connect as a production-grade SaaS application.

18

EXTERNAL APIS

5

CLOUD PROVIDERS

4

BUDGET TIERS

21

API ROUTES

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 Quick Budget Overview

MVP / DEMO

\$8

₹660/mo

0 - 500 users

SMALL PRODUCTION

\$70

₹5,880/mo

500 - 2K users

MEDIUM SCALE

\$310

₹26,000/mo

2K - 10K users

LARGE SCALE

\$950

₹79,800/mo

10K - 50K users

Tech Stack Summary

LAYER	TECHNOLOGY	PURPOSE
Frontend	React 18 + Tailwind CSS + Three.js + Framer Motion	SPA with 3D visuals, animations
Backend	Node.js + Express.js	REST API server (21 route modules)
Database	MongoDB (Atlas)	Primary data store (23 models)
Cache/PubSub	Redis	Caching + Socket.IO adapter
Real-time	Socket.IO (Redis adapter)	Live chat, notifications
File Storage	Cloudinary	Image/file uploads & CDN
Email	Brevo (Sendinblue)	OTP, notifications, weekly digests
SMS	Fast2SMS	OTP & notification forwarding (India)
AI Primary	Groq API (compound-mini)	Career mentor, resume, interviews
AI Fallback	Google Gemini (3.x Flash)	Fallback AI engine
Auth	Google OAuth + GitHub OAuth + JWT	Multi-provider authentication
Code Exec	Judge0 CE (RapidAPI)	Coding contest evaluation (planned)
Dev Stats	GitHub + LeetCode + HackerRank + GFG + CodeChef + CF + Duolingo	Developer profile aggregation
Mobile	Capacitor (Android)	Mobile APK wrapper

API Routes (21 Modules)

```
/api/auth → Authentication (Google, GitHub, JWT, OTP, 2FA) /api/users → User profiles, search, connections
/api/mentorship → Mentorship requests, matching, sessions /api/jobs → Job listings, applications, AI matching
/api/events → Event creation, RSVP, calendar /api/forum → Discussion posts, comments, voting /api/contests → Coding
contests, submissions (Judge0) /api/messages → Real-time chat (Socket.IO), group chats /api/admin → Admin dashboard,
user management /api/notifications → Push notifications system /api/upload → File uploads (Cloudinary) /api/dev-
activity → Developer stats aggregation /api/leaderboard → Gamification leaderboard /api/feedback → User feedback
collection /api/helpdesk → Support ticket system /api/projects → Project showcase /api/institutions →
College/institution management /api/tasks → Task management /api/ai → AI Career Mentor (Chat, Resume, Interview)
/api/tech-hub → Tech news/resources /api/business → Business directory
```

MongoDB Models (23)

```
User, Message, ChatGroup, Contest, DevProfile, Event, Feedback, ForumPost, HelpDesk, Job, JobApplication,
JobOpportunity, MentorReview, MentorReward, Mentorship, MentorshipRequest, MentorshipSession, Notification, Project,
Report, Task, TechHub, Business
```

Component	Minimum	Recommended
Backend Server	1 vCPU, 1 GB RAM	2 vCPU, 2 GB RAM
MongoDB	512 MB (Atlas M0 Free)	M10 Dedicated (2 GB RAM)
Redis	256 MB	1 GB (Socket.IO + caching)
Frontend Hosting	Static site CDN	Static site CDN
Domain	1x custom domain	.com or .in domain
SSL Certificate	Required (HTTPS)	Free with most hosts
CI/CD Pipeline	GitHub Actions (free)	GitHub Actions
Monitoring	Basic health checks	APM + error tracking

#	SERVICE	PURPOSE	AUTH	SOURCE FILE
1	Groq API	AI Chat, Resume, Mock Interview, Job Match	API Key	aiHelper.js, ai.js
2	Google Gemini	AI Fallback Engine	API Key	aiHelper.js
3	Brevo (Sendinblue)	Transactional Emails (OTP, Welcome, Digest)	API Key	sendEmail.js
4	Fast2SMS	SMS OTP & Notifications (India)	API Key	sendSMS.js
5	Cloudinary	Image/File Upload & CDN	Key + Secret	cloudinary.js
6	Google OAuth 2.0	Google Sign-In	Client ID	authService.js
7	GitHub OAuth	GitHub Sign-In	ID + Secret	authService.js
8	GitHub GraphQL	Dev Stats (repos, contributions, badges)	PAT	devStatsFetcher.js
9	LeetCode GraphQL	Coding stats, badges, ratings	Public	devStatsFetcher.js
10	HackerRank REST	Profile, badges, certificates	Public	devStatsFetcher.js
11	GFG Scraper	Coding score, problems solved	Scraping	devStatsFetcher.js
12	CodeChef Scraper	Rating, stars, division	Scraping	devStatsFetcher.js
13	Codeforces API	Rating, contests, problems	Public	devStatsFetcher.js
14	Duolingo API	Language learning stats	Public	devStatsFetcher.js
15	Disify API	Email validation (disposable detection)	Public	authService.js
16	JudgeO CE	Code execution for contests	API Key	contests.js (TODO)
17	Google Userinfo	OAuth profile data	Bearer	authService.js
18	GitHub User API	OAuth email retrieval	Bearer	authService.js

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Cloud Hosting Provider Comparison

Backend Server (Node.js + Express + Socket.IO)

Requirement: 1-2 vCPU, 1-2 GB RAM, always-on, WebSocket support, auto-deploy from GitHub

FEATURE	RENDER	AWS	AZURE	DIGITALOCEAN	RAILWAY
Starter Plan	\$7/mo	~\$15-18/mo	~\$13/mo	\$6/mo	\$5/mo
Production Plan	\$25/mo	~\$18/mo	~\$13/mo	\$12/mo	\$20/mo
WebSocket	✔ Native	✔ With ALB	✔ Config	✔ Native	✔ Native
Auto-Deploy	✔ Built-in	⚠ CodePipeline	⚠ DevOps	✔ App Platform	✔ Built-in
Free SSL	✔	⚠ ACM+ALB	✔	✔	✔
Scaling	Manual/Auto	Auto Groups	Scale-up/out	Manual	Auto
Cold Start	⚠ Free sleeps	✖ None	✖ None	✖ None	⚠ Starter
Hidden Costs	None	EBS, ALB (\$20+)	Data egress	BW overages	Bandwidth
Ease of Setup	★★★★★	★★	★★★	★★★★★	★★★★★
Best For	Small-Med	Enterprise	Microsoft	Budget Prod	Quick Deploy

Frontend Hosting (React Static Build) — All Free Options

PROVIDER	COST	CDN	BANDWIDTH	NOTES
Cloudflare Pages	FREE	✔ Fastest	Unlimited	★ Recommended
Vercel	FREE	✔ Edge	100 GB/mo	Best React DX
Netlify	FREE	✔ CDN	100 GB/mo	Great CI/CD
Render Static	FREE	✔ Global	100 GB/mo	Current setup
Azure Static Web Apps	FREE	✔ Azure	100 GB/mo	2 free apps

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Database Tier Comparison

PROVIDER	FREE TIER	COST/MO	STORAGE	RAM	NOTES
MongoDB Atlas M0	✔ 512 MB	\$0	512 MB	Shared	Current. 100 conn max
MongoDB Atlas M10	✗	~\$57	10 GB	2 GB	Recommended starter prod
MongoDB Atlas M20	✗	~\$146	20 GB	4 GB	Medium scale
AWS DocumentDB	✗	~\$200+	Pay/GB	Varies	Expensive
Azure Cosmos DB	✔ 1000 RU/s	~\$25+	Pay/GB	Varies	Complex RU pricing
DigitalOcean MongoDB	✗	\$15	10 GB	1 GB	Simple & affordable
Self-hosted (VM)	✗	\$0 extra	Disk	VM	You manage it

Scale-Based Recommendations

SCALE	USERS	EST. DB SIZE	TIER	COST/MO
MVP / Demo	0 - 500	< 200 MB	Atlas M0 (Free)	\$0
Single College	500 - 2K	200 MB - 1 GB	Atlas M0 or Shared	\$0 - \$9
Multi-College	2K - 10K	1 - 5 GB	Atlas M10	~\$57
Regional	10K - 50K	5 - 20 GB	Atlas M20	~\$146
National	50K+	20+ GB	Atlas M30+	\$300+

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Redis / Caching Tier Comparison

Used for: API response caching (30-min TTL) + Socket.IO adapter (multi-instance WebSocket scaling)

PROVIDER	FREE TIER	COST/MO	MEMORY	NOTES
Upstash Redis	✔ 256 MB, 10K/day	\$0 → \$10+	256 MB	★ Serverless, best free
Redis Cloud	✔ 30 MB	\$0 → \$5+	30-250 MB	Official Redis
Render Key Value	✔ 25 MB	\$0 → \$10	25-256 MB	Same ecosystem
AWS ElastiCache	✗	~\$13	0.5 GB	Enterprise-grade
Azure Cache	✗	~\$16	250 MB	Azure ecosystem

7.1 — Groq API (Primary AI Engine)

Used for: AI Career Mentor, Resume Analysis, Mock Interview, Job Matching, Interview Evaluation

Component	Pricing
Free Tier	✔ Available (rate-limited RPM/TPM per org)
Model Tokens (compound-mini)	~\$0.15 / 1M input, ~\$0.60 / 1M output
Web Search (if used)	\$5 - \$8 / 1,000 requests
Code Execution	\$0.18 / hour

Estimated Monthly Cost by Scale

Scale	AI Requests/Mo	Est. Tokens/Mo	Monthly Cost
MVP (<100 users)	~500	~2M	\$0 (Free)
Small (500 users)	~2,000	~10M	\$2 - \$5
Medium (5K users)	~15,000	~75M	\$15 - \$40
Large (25K users)	~75,000	~375M	\$75 - \$200

7.2 — Google Gemini API (AI Fallback)

**Action Required**

Your code uses `gemini-2.0-flash` which was retired in June 2026. Update `aiHelper.js` to use `gemini-3.5-flash` or `gemini-3.7-flash`.

Component	Pricing
Free Tier	✔ Available via Google AI Studio (rate-limited)
Gemini 3.7 Flash	\$0.75 / 1M input tokens, \$3.75 / 1M output tokens

Scale	Fallback Requests/Mo	Monthly Cost
MVP	~50	\$0
Small	~200	\$0 - \$1
Medium	~1,500	\$2 - \$8
Large	~7,500	\$15 - \$50

7.3 — Brevo (Sendinblue) — Transactional Emails

Used for: OTP verification, welcome emails, password resets, peer notifications, weekly Dev Pulse digests

PLAN	COST	LIMIT	BRANDING
Free	\$0/mo	300 emails/day (~9K/mo)	"Sent with Brevo"
Starter	\$9/mo	5,000 emails/mo	Watermark (+\$12 remove)
Business	\$18/mo	5,000 emails/mo	No watermark

SCALE	OTP	NOTIFICATIONS	DIGEST	TOTAL/MO	COST
MVP (100 users)	~200	~300	~400	~900	\$0
Small (500)	~800	~1,500	~2,000	~4,300	\$0
Medium (5K)	~3K	~10K	~20K	~33K	\$25/mo
Large (25K)	~10K	~50K	~100K	~160K	\$65/mo

7.4 — Fast2SMS — SMS Notifications (India Only)

WALLET AMOUNT	PER SMS COST	NOTES
₹100 - ₹3,999	₹0.25	+ GST
₹4,000 - ₹7,999	₹0.21	+ GST
₹8,000 - ₹13,999	₹0.19	+ GST
₹14,000+	₹0.17	+ GST
Quick SMS (No DLT)	₹5.00	International route

SCALE	SMS/MO	COST (₹0.25)	COST (₹0.17)
MVP	~100	₹25 (~\$0.30)	₹17
Small	~500	₹125 (~\$1.50)	₹85
Medium	~3,000	₹750 (~\$9)	₹510
Large	~15,000	₹3,750 (~\$45)	₹2,550

7.5 — Cloudinary — File Uploads & Image CDN

PLAN	COST	CREDITS	STORAGE	BANDWIDTH
Free	\$0/mo	25	25 GB	25 GB
Plus	\$89/mo	225	225 GB	225 GB
Advanced	\$249/mo	1,000	1 TB	1 TB

7.6 — Judge0 CE (RapidAPI) — Code Execution

PLAN	COST	SUBMISSIONS
Basic (Free)	\$0/mo	~50/day (shared, unreliable)
Pay-Per-Use	~\$0.0017/submission	Unlimited
Self-Hosted (Docker)	VM cost only	Unlimited

7.7 — Free APIs (No Cost)

SERVICE	COST	NOTES
Google OAuth 2.0	FREE	Unlimited. Only basic scopes needed.
GitHub OAuth	FREE	Unlimited.
GitHub GraphQL API	FREE	5,000 points/hr with PAT
LeetCode GraphQL	FREE	Unofficial, may break

SERVICE	COST	NOTES
HackerRank REST	FREE	Unofficial, may break
Codeforces API	FREE	1 req/2 sec rate limit
CodeChef (Scraping)	FREE	May break without notice
GFG (Scraping)	FREE	May break without notice
Duolingo API	FREE	Unofficial
Disify (Email Validation)	FREE	10K validations/day (free acct)

ITEM	PROVIDER	COST	NOTES
alumnexconnect.com	Namecheap	~\$7 (1st yr) → \$18/yr renewal	Recommended
alumnexconnect.in	Namecheap / GoDaddy	~\$10-11/yr	India-specific
SSL Certificate	Cloudflare / Let's Encrypt	FREE	Auto-provisioned
CDN	Cloudflare Free	FREE	Unlimited BW, best free CDN
DNS	Cloudflare	FREE	Fast DNS + DDoS protection



SECURITY ALERT

Your .env file contains **real API keys, database credentials, and OAuth secrets**. Rotate ALL keys immediately before going public. Never commit .env to Git.

#	TASK	STATUS	PRIORITY
1	Rotate ALL API keys & secrets	CRITICAL	P0
2	Set strong JWT_SECRET (currently placeholder)	CRITICAL	P0
3	Update Gemini model to gemini-3.x	Required	P1
4	Remove self-ping hack (Render free tier workaround)	Production	P2
5	Add error monitoring (Sentry free: 5K events/mo)	Required	P1
6	Add health check endpoint (/api/health)	Required	P1
7	Enable MongoDB Atlas IP allowlist	Required	P1
8	Register DLT for production SMS (India requirement)	Required	P1
9	Google OAuth consent screen verification (>100 users)	Required	P1
10	Implement Judge0 integration (currently TODO)	Nice to have	P2
11	Add rate limiting for AI endpoints (prevent abuse)	Required	P1
12	Enable gzip/brotli compression	Done	—
13	Set up CI/CD pipeline (GitHub Actions)	Recommended	P2
14	Add Redis connection pooling config	Optimization	P2

#	TASK	STATUS	PRIORITY
15	Configure structured production logging	Recommended	P2

10.1 — MVP / College Demo (0-500 users)

SERVICE	PROVIDER	PLAN	USD/MO	INR/MO
Backend Server	Render	Starter	\$7	₹588
Frontend Hosting	Cloudflare Pages	Free	\$0	₹0
Database	MongoDB Atlas	M0 Free	\$0	₹0
Redis/Cache	Upstash	Free	\$0	₹0
Email	Brevo	Free (300/day)	\$0	₹0
SMS	Fast2SMS	~100 SMS	~\$0.30	~₹25
AI (Groq)	Groq	Free Tier	\$0	₹0
AI (Gemini)	Google	Free Tier	\$0	₹0
File Uploads	Cloudinary	Free	\$0	₹0
Domain	Namecheap	.com	~\$0.58	~₹50
SSL + CDN	Cloudflare	Free	\$0	₹0
Monitoring	Sentry	Free	\$0	₹0
TOTAL (Monthly)			~\$8	~₹660

10.2 — Small Production (500-2,000 users)

SERVICE	PROVIDER	PLAN	USD/MO	INR/MO
Backend	Render	Standard	\$25	₹2,100
Workspace	Render	Pro	\$25	₹2,100
Frontend	Cloudflare Pages	Free	\$0	₹0
Database	MongoDB Atlas	Shared	\$9	₹756
Redis	Upstash	Pay-as-you-go	~\$5	₹420
Email	Brevo	Free	\$0	₹0
SMS	Fast2SMS	~500 SMS	~\$1.50	~₹125
AI	Groq	Pay-as-you-go	~\$3	₹252
Domain	Namecheap	.com renewal	~\$1.50	~₹126
TOTAL (Monthly)			~\$70	~₹5,880

10.3 — Medium Production (2,000-10,000 users)

SERVICE	PROVIDER	PLAN	USD/MO	INR/MO
Backend	Render / AWS	Production	\$25	₹2,100
Workspace	Render Pro	—	\$25	₹2,100
Frontend	Cloudflare Pages	Free	\$0	₹0
Database	MongoDB Atlas	M10 Dedicated	\$57	₹4,788
Redis	Redis Cloud	Essentials	\$5	₹420
Email	Brevo	Starter (20K)	\$25	₹2,100
SMS	Fast2SMS	~3K SMS	~\$9	~₹750
AI (Groq)	Groq	Pay-as-you-go	~\$25	₹2,100
AI (Gemini)	Google	Pay-as-you-go	~\$5	₹420
File Uploads	Cloudinary	Plus	\$89	₹7,476
Judge0	RapidAPI	Pay-per-use	~\$17	₹1,428
Domain	Namecheap	.com	~\$1.50	~₹126
Monitoring	Sentry	Team	\$26	₹2,184
TOTAL (Monthly)			~\$310	~₹26,000

10.4 — Large Scale (10,000-50,000 users)

SERVICE	PROVIDER	PLAN	USD/MO	INR/MO
Backend (2x)	AWS ECS / Azure	2x instances	\$60	₹5,040
Load Balancer	AWS ALB	—	\$20	₹1,680
Frontend	Cloudflare Pages	Pro	\$20	₹1,680
Database	MongoDB Atlas	M20 Dedicated	\$146	₹12,264
Redis	Redis Cloud	1 GB	\$32	₹2,688
Email	Brevo	Business (60K)	\$65	₹5,460
SMS	Fast2SMS	~15K SMS	~\$45	~₹3,750
AI (Groq)	Groq	Pay-as-you-go	~\$150	₹12,600
AI (Gemini)	Google	Pay-as-you-go	~\$35	₹2,940
File Uploads	Cloudinary	Advanced	\$249	₹20,916
Judge0	Self-hosted	Dedicated VM	\$25	₹2,100
Monitoring	Sentry + Datadog	Team	\$50	₹4,200

SERVICE	PROVIDER	PLAN	USD/MO	INR/MO
Backup/DR	Atlas + S3	—	\$30	₹2,520
Domain + CDN	Cloudflare Pro	—	~\$21.50	~₹1,800
TOTAL (Monthly)			~\$950	~₹79,800

10.5 — Provider Stack Comparison (Medium Scale Total Monthly Cost)



STACK	BACKEND	DB	REDIS	EMAIL	CDN	TOTAL/MO
DigitalOcean	\$12	\$15	\$10	\$25	\$0	~\$200
Railway	\$20	\$57	\$5	\$25	\$0	~\$245
Azure	\$13	\$25	\$16	\$25	\$0	~\$265
Render	\$50	\$57	\$10	\$25	\$0	~\$280
AWS	\$38	\$57	\$13	\$25	\$3	~\$320

DigitalOcean

\$200

🏆 Winner: Cost

Railway

\$245

Easy + Affordable

Azure

\$265

Microsoft Ecosystem

Render

\$280

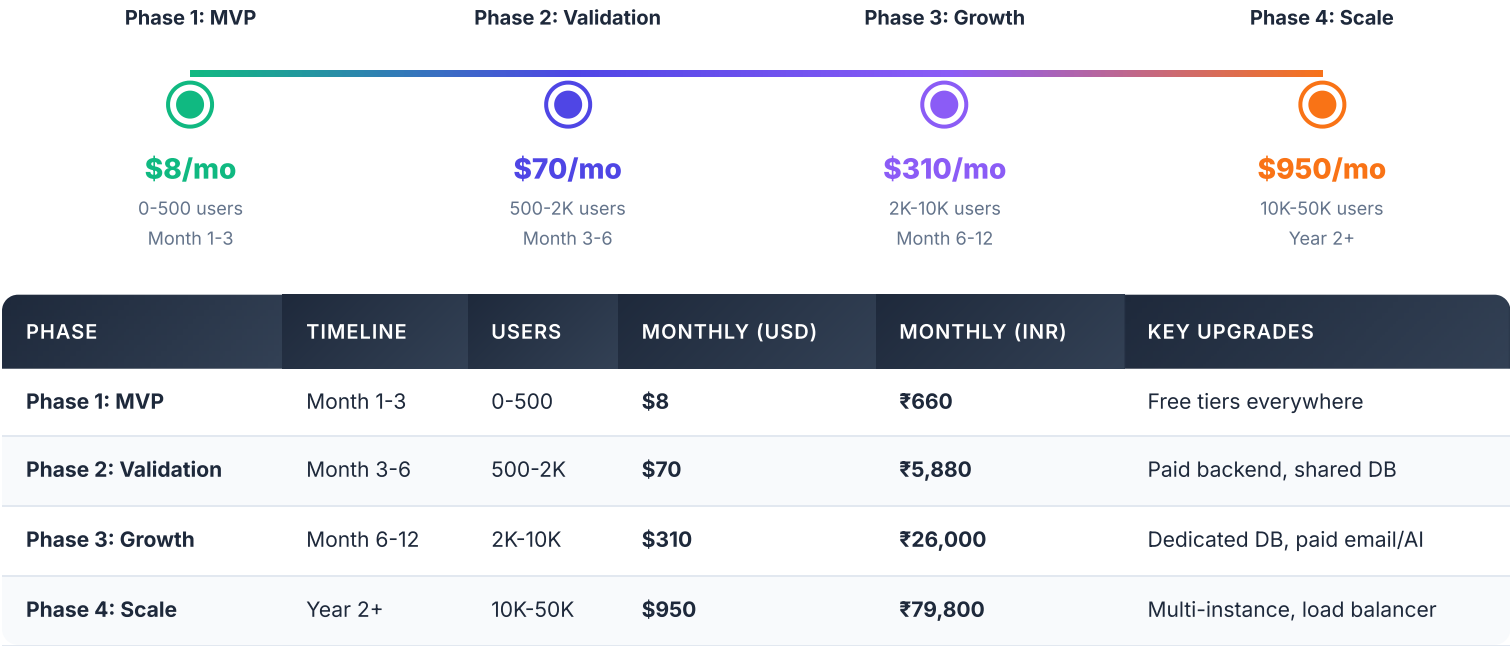
🏆 Winner: Ease

AWS

\$320

🏆 Winner: Scale

11 Scaling Roadmap & Cost Projections



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Recommended Production Stack



Best Value Stack for Launch

This stack uses free tiers strategically to minimize cost while maintaining production quality.

COMPONENT	PROVIDER	PLAN	COST/MO
Backend	Render Recommended	Starter → Standard	\$7 - \$25
Frontend	Cloudflare Pages Recommended	Free	\$0
Database	MongoDB Atlas	M0 Free → M10	\$0 - \$57
Redis	Upstash Recommended	Free → Pay-as-you-go	\$0 - \$5
Email	Brevo	Free (300/day)	\$0
SMS	Fast2SMS	Wallet	~\$1
AI	Groq	Free → Pay-as-you-go	\$0 - \$5
CDN + DNS + SSL	Cloudflare	Free	\$0
Domain	Namecheap	.com	~\$1/mo
Monitoring	Sentry	Free	\$0
TOTAL RANGE			\$8 - \$94/mo

⚡ Immediate Action Items to Go Live

- 1 Buy domain → [alumnexconnect.com](#) on Namecheap (~\$7)
- 2 Set up Cloudflare → Point domain, enable free CDN + SSL
- 3 Deploy backend → Render Starter (\$7/mo) with all env vars
- 4 Deploy frontend → Cloudflare Pages (free) with REACT_APP_API_URL
- 5 Rotate all keys → Generate fresh secrets for all APIs
- 6 Google OAuth verification → Submit consent screen for >100 users
- 7 DLT registration → Register for production SMS (if needed)
- 8 Update Gemini model → Change to gemini-3.7-flash in aiHelper.js

9 **Test everything** → Run through all features on production

10 **Set up monitoring** → Free Sentry account for error tracking

💡 **Bottom Line**

You can launch Alumnex Connect as a production SaaS for **as low as ₹660/month (\$8/mo)** using free tiers strategically. As you grow to 10,000+ users, expect costs to reach **₹26,000/month (~\$310/mo)**. The platform is architecturally sound for scaling — the main cost drivers at scale are **Cloudinary** (file storage), **MongoDB Atlas** (database), and **AI APIs** (Groq/Gemini).